

Session 10 (AFRM):

Conditional:-

$$\sigma_{t+1}^2 = w + \alpha \overset{\text{MA}}{R_t^2} + \beta \overset{\text{AR}}{\sigma_t^2}$$

σ_{t+1}^2 is conditional on R_t^2 & σ_t^2

or

$$\sigma_{t+1}^2 \mid \sigma_t^2, R_t^2 = w + \alpha R_t^2 + \beta \sigma_t^2$$

Implementing GARCH model in R:

We need rugarch library. Install "rugarch" package once.

To implement GARCH model in R, we follow two steps process.

First, we need to specify GARCH model using "rugarchspec" command.

Spec1

Variance model	mean model	distribution model
→ First mention the type of GARCH model to use:- sGARCH, gjrGARCH, eGARCH → Next, we need to mention the order of GARCH model:- $c(1, 1)$ $\swarrow$ $\searrow$ σ_t^2 R_t^2 (AR) (MA)	→ For VAR estimation, we don't need to consider model, i.e. mean model is $R_{t+1} = \epsilon_{t+1}$ → armaOrder = c(0, 0) We will also not consider intercept. ∴ include.mean = FALSE	norm = Normal std = Student t distribution ged = Generalized Error distribution snorm = Skewed Normal sstd = Skewed Student t sged = Skewed GED

Next, we need to fit GARCH model, using "rugarchfit" command.

result = rugarchfit(spec = spec1, data = return series)

Any variable name

- Once GARCH model is fitted, we can extract fitted volatilities for backtesting.
- Next, we can forecast volatilities using "ugarchforecast" command.
- `ugarchforecast (name of fitted model, n.ahead=1)`
- this will generate next day forecast of volatility.

Notes:-

$$1. \quad \sigma_{t+1}^2 = w + \alpha R_t^2 + \beta \sigma_t^2$$

$$= w + \alpha R_t^2 + \beta (w + \alpha R_{t-1}^2 + \beta \sigma_{t-1}^2)$$

$$= w + \beta w + \alpha R_t^2 + \beta \alpha R_{t-1}^2 + \beta \cdot \beta \sigma_{t-1}^2$$

GARCH order = (1, 1)

AR σ_t^2 MA R_t^2

② Identify the Significance of Regression Coeff

We can look at p-value of the Coeff. to know if given independent variable is significantly impacting dependent variable. ($Y = \alpha + \beta X + \epsilon$)

dependent variable

independent variable

Look at p-value of β to conclude if X significantly impacts Y .

We consider three different levels of significance.

① 10% level of significance:

If p -value of coeff. is $\leq 10\%$ & $> 5\%$, this indicates that the given X variable impacts Y variable at 10% level of significance. This indicates that there is weaker evidence that X impacts Y .

② 5% level of significance:

If p -value of coeff. is $\leq 5\%$ & $> 1\%$, then the given X -variable is significantly impacting Y -variable at 5% level of significance. This indicates that there is a moderate evidence that X -variable impacts Y -variable.

③ 1% level of significance:

If p -value is $\leq 1\%$, then the result is significant at 1% level of significance. That is, there is strong evidence that X impacts Y .

If p -value $> 10\%$ $\Rightarrow$ Result is insignificant

— X — X — X — X —
GJR (Glosten - Jagannathan - Runkle) GARCH model: This model incorporates leverage effect while modelling variance.

Leverage effect indicates that -ve news increases the volatility by larger magnitude than +ve news of similar intensity.

Fundamentally, we refer $\frac{D}{E}$ ratio by leverage.

When there is -ve news in a market, the Equity (E) value fall. This means $\left(\frac{D}{E}\right)$ will appear to increase.

The increase in $\frac{D}{E}$ ratio further increases

the default risk of firm which further increases the risk of equity holder which further increases the volatility of Equity.

In GJR-GARCH model,
mean eq.:

$$R_{t+1} = \epsilon_{t+1}$$

Variance eq.:

$$\sigma_{t+1}^2 = \omega + \alpha R_t^2 + \beta \sigma_t^2 + \gamma \cdot I_{\{R_t < 0\}} \cdot R_t^2$$

where $I_{\{R_t < 0\}} = \begin{matrix} \text{indicator variable} \\ \text{with condition} \\ R_t < 0 \end{matrix} = \begin{cases} 1, & \text{when } R_t < 0 \\ 0, & \text{otherwise} \end{cases}$

Case 1: if $R_t < 0$, then $I_{\{R_t < 0\}} = 1$

$$\therefore \sigma_{t+1}^2 = \omega + \alpha R_t^2 + \beta \sigma_t^2 + \gamma \cdot 1 \cdot R_t^2$$

$$\sigma_{t+1}^2 = \omega + (\alpha + \gamma) R_t^2 + \beta \sigma_t^2 > \text{GARCH Variance}$$

i.e. increased variance if $R_t < 0$

Case 2: if $R_t \geq 0$, $I_{\{R_t < 0\}} = 0$

$$\sigma_{t+1}^2 = \omega + \alpha R_t^2 + \beta \sigma_t^2 + \underbrace{\gamma \cdot 0 \cdot R_t^2}_{= 0}$$

$$\sigma_{t+1}^2 = \omega + \alpha R_t^2 + \beta \sigma_t^2 = \text{GARCH Variance.}$$

In R, use "gjrGARCH" as model name.

Notation:

$$r_t \sim N(0, \sigma^2)$$

$$r_t \sim \sigma \cdot Z \rightarrow \text{Standard Normal}$$

$$r_t = \varepsilon_t \sim \sigma Z$$

$$\frac{r_t}{\sigma} \text{ or } \frac{\varepsilon_t}{\sigma} = Z \rightarrow 5\% = -1.645$$

EGARCH (Exponential GARCH) model:

It also captures leverage & model $\ln(\sigma_{t+1}^2)$.

mean eq:

$$R_{t+1} = \varepsilon_{t+1}$$

Variance eq:

$$\ln(\sigma_{t+1}^2) = \omega + \alpha \left[\frac{|R_t|}{\sigma_t} - \sqrt{\frac{2}{\pi}} \right] + \beta \cdot \ln(\sigma_t^2)$$

$$+ \gamma \cdot \frac{R_t}{\sigma_t}$$

Here, if gamma is -ve & significant then we say there is a leverage effect.

In R, use "eGARCH" in ugarchspec.

